

EV Market Sales Analysis

Trends & Market Analytical Brief

1. The Business Challenge :

The global EV market (2,000 models across 2020–2026, spanning 6 countries) is rapidly maturing, but purchasing behaviour across segments, geographies, and vehicle types is shifting faster than most players can track. Without a clear analytical view of what's driving sales, pricing power, and customer satisfaction, businesses risk misallocating R&D and marketing spend, ceding ground to data-savvy competitors like Tesla and BYD.

Core Business Question: *"How can we use EV market data to identify which segments, brands, and product features drive sales volume, revenue, and customer loyalty — and where the biggest untapped opportunities lie?"*

2. Why This Matters :

Business Area	Whats Stake
Revenue Growth	Luxury and Premium segments already generate \$8.3T and \$7.1T in combined market revenue — understanding what drives them is critical
Customer Retention	Average customer rating sits at just 3.57/5 — there's a clear satisfaction gap to close
Marketing Efficiency	Tesla dominates with 101M units sold; understanding their positioning vs. laggards reveals what messaging works
Product Strategy	Range averages only 236 miles in Budget vs. 275 in Luxury — segment-level feature gaps need quantifying
Competitive Advantage	US and German manufacturers hold 65% of model share; South Korean and Chinese brands are closing fast

3. Key Questions This Project Will Answer :

- Which market segments (Budget / Mid-range / Premium / Luxury) generate the best balance of volume and revenue?
- Do higher charging speeds and autopilot levels translate into measurably better customer ratings and sales?
- How do pricing bands (\$16K–\$270K) affect sales velocity across body types (SUV, Truck, Sedan, etc.)?
- Which countries of origin are gaining or losing market share — and why?
- What role do safety ratings and warranty length play in purchasing decisions?
- Which brands outside Tesla represent the highest-growth opportunity?

4. Project Scope & Deliverables :

Mirror the four workstreams from the reference brief:

1. **Data Preparation (Python)** — Clean and engineer features: price tiers, range bands, autopilot adoption rate, revenue per model
2. **Data Analysis (SQL)** — Segment by country, body type, drive type, and year to surface trend patterns
3. **Interactive Dashboard (Power BI / Tableau)** — Filterable by brand, segment, year, and geography for non-technical stakeholders
4. **Report & Presentation** — Strategic recommendations on segment investment, feature prioritisation, and competitive positioning
5. **GitHub Repository** — Reproducible codebase for ongoing market tracking

5. Methodology :

- **Data Audit:** 2,000 records, 24 variables — validate nulls, outliers (e.g., \$270K ceiling prices), and cross-field consistency (e.g., autopilot level vs. price tier)
- **Feature Engineering:** Revenue per model (price × sales), performance-to-price ratio, customer satisfaction index by segment
- **Segmentation Analysis:** Cluster models by price, range, and sales to reveal natural market groupings
- **Correlation Analysis:** Identify which specs (charging speed, safety rating, autopilot level) most strongly predict sales and ratings

6. Expected Outcomes :

- A clear view of which segments and brands to prioritise for investment or partnership
- Feature benchmarks by segment to guide product roadmap decisions
- Identification of underperforming models with high price but low ratings — churn risk indicators
- A reusable dashboard for ongoing EV market monitoring as new model years enter the dataset

7. Audience :

Business Stakeholders	Technical Practitioners
Dashboard and presentation in plain language — which segments win, which brands to watch, what customers want	Full Python cleaning scripts, SQL queries, and documented methodology in GitHub for reproducibility